

SUBJECT NO-MA20203, SUBJECT NAME- PARTIAL DIFFERENTIAL EQUATIONS LTP- 3-0-0, CRD- 3

SYLLABUS :- Part A: First-Order Partial Differential Equations(PDEs) : Introduction to PDEs: Why PDEs? Application to real life problems; curves and surfaces; ideas on order, degree, linear, non-linear; formation and classification of first order PDEs; Classification of integrals as general solution, complete integral and singular solution. Lagrange's method of solution for quasi-linear PDEs; (The method of characteristics); Solution of Cauchy problem for first order quasi-linear pdes (with and without using general integral); Pfaffian differential equation, Compatible system, Charpit's method, and Jacobi's method for nonlinear PDEs; Cauchy's problem for first order Non-linear PDEs;

Part B: Second-Order PDEs B1. Second and higher order Linear PDEs with constant coefficients; Finding complementary function (CF) and particular integral (PI); Special forms of PI. B2. Classification of second order linear PDE; Characteristics; Canonical forms; Well-posed problems. B3. Hyperbolic equation: Derivation of the wave equation; The infinite string problem; The D'Alembert solution of the wave equation; The semi-infinite string problem. B4. Hyperbolic equation: Basics of Fourier series; Wave equation; The finite vibrating string problem; The method of separation variables. B5. Parabolic equation: 1-dimensional heat conduction equation, separation of variable solution for homogeneous equations; Time independent and dependent boundary conditions. B6. Elliptic equation: 2-dimensional Laplace equation; Classification of boundary value problems with respect to Dirichlet; Neumann and Robin boundary conditions; The maximum and minimum principle; Separation of variable solution for (i) Dirichlet problem for a rectangle; (ii) Neumann problem for a rectangle; (iii) Interior and Exterior Dirichlet problems for a circle; (iv) Dirichlet problem in cylindrical and spherical coordinates. B7. Duhamel's principle to obtain a solution for non-homogeneous equations.

SL No	Topics	Lectures
1.	Introduction to PDEs: Why PDEs? Application to real life problems	1
2.	curves and surfaces; ideas on order, degree, linear, non-linear; formation and classification of first order PDEs	3
3.	Classification of integrals as general solution, complete integral and singular solution	1
4.	Lagrange's method of solution for quasi-linear PDEs	1
5.	Solution of Cauchy problem for first order quasi-linear pdes (with and without using general integral)	3
6.	Pfaffian differential equation, Compatible system, Charpit's method; Jacobi's method for nonlinear PDEs;	5
7.	Cauchy's problem for first order Non-linear PDEs	2
8.	Second and higher order Linear PDEs with constant coefficients; Finding complementary function (CF) and particular integral (PI); Special forms of PI	3
9.	Classification of second order linear PDE; Characteristics; Canonical forms; Well-posed problems	2
10.	Derivation of the wave equation; The infinite string problem; The D'Alembert solution of the wave equation; The semi-infinite string problem	3
11.	Basics of Fourier series; Wave equation; The finite vibrating string problem; The method of separation variables	4
12.	1-dimensional heat conduction equation, separation of variable solution for homogeneous equations; Time independent and dependent boundary conditions.	3

13.	2- dimensional Laplace equation; Classification of boundary value problems with respect to Dirichlet; Neumann and Robin boundary conditions; The maximum and minimum principle	2
14.	Separation of variable solution for (i) Dirichlet problem for a rectangle; (ii) Neumann problem for a rectangle; (iii) Interior and Exterior Dirichlet problems for a circle; (iv) Dirichlet problem in cylindrical and spherical coordinates	5
15.	Duhamel's principle to obtain a solution for non-homogeneous equations	2
Total		40

Texts and References:

1. An Elementary course in Partial Differential Equations – T.Amarnath, Jones and Bartlett Learning (Foreign Ed.) / Narosa (Indian Ed.).
2. Elements of Partial Differential Equations- Ian N Sneddon, Dover Publications.
3. Linear Partial Differential Equations for Scientists and Engineers – Tyn Myint-Uand Lokenath Debnath, Birkhauser
4. Elementary Applied Partial Differential Equations - R.Haberman, Prentice Hall
Differential Equations - R.Haberman, Prentice Hal